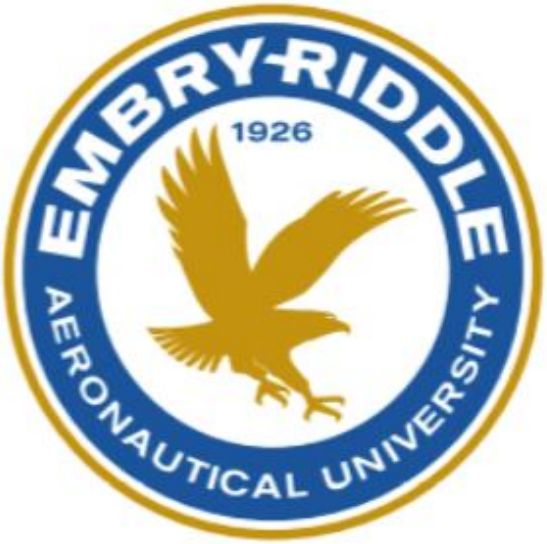



# LUNAR LYFT

CONCEPTUAL DESIGN REPORT

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October 15, 2026

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# Executive Summary

## Introduction

Demands by the government and private sector for human presence on the moon continue to grow over time. The call for cislunar operations includes tourism, research, exploration, mining, and manufacturing. Assuming sufficient investment, Lunar Lyft has been contracted to create transportation architecture to carry sufficient crew and cargo (including scientific and recreational equipment) for a cislunar space mission. The design of lunar habitation systems shall also be conceptualized.

## Problem statement

The mission must take passengers from an earth space station to the moon in a structurally reliable launch vehicle using proper staging. The destination must include at least three lunar habitation facilities that can support life for long-term research, manufacturing, and commercial usage.

## Summary of Selected Design

The rocket configuration includes a three-stage rocket carrying 20 people – including crew, supplies and emergency supplies for both the people onboard and in lunar orbit. Lunar habitations include a main station and hotel in Near Rectilinear Halo Orbit (NRHO), and a mining and research facility on the lunar south pole.

## CONOPS: Assumptions, Mission Requirements & Design Features/Performance Capabilities

Assumptions for this mission include:

- Structures on the surface and around the Moon's orbit are already built.
- Stage 1 and Stage 3 reusable and refuellable
- All parts of the mission can support safety requirements.
- Unlimited water can be mined on the Moon's poles.
- Resources mined from the Moon are sufficient for our propulsion systems.
- Our payload is comprised of food and supplies for…
    - 20 people (including the operating crew)
    - Cislunar Orbital Station
    - Lunar Habitats
    - Emergency situations

The Lunar Lyft Mission has been designed to transport up to twenty people, supplies, and cargo from Earth to the Moon. Lunar Lyft operates as a 3-stage rocket where stage 1 provides enough thrust to leave earth's atmosphere from a sea-level launch. Stage 1 is fully recoverable and will return to earth after initial separation from stage 2 and stage 3. Stage 1 will then proceed to be refueled and reused in future missions. After separation, once the vehicle reaches LEO, Stage 2 initiates separation and remains in orbit. The second stage is not refueled or recovered, allowing. Stage 3 proceeds to the moon by initiating a Translunar injection. Stage 3 then proceeds to enter NHRO which allows for rendezvous and docking with COLOSAL.

COLOSAL allows for lunar elevator access which can provide further access to the lunar south pole. Crewed missions to the Nobile rim area in the lunar south pole would then allow for research, water access, and mining capabilities. Lunar resources allow for sustaining future missions by providing water for mission critical operations and the ability to manufacture fuel. After completion of lunar operations, Stage 3 is refueled and returns to earth where it is recovered and reused in future missions.

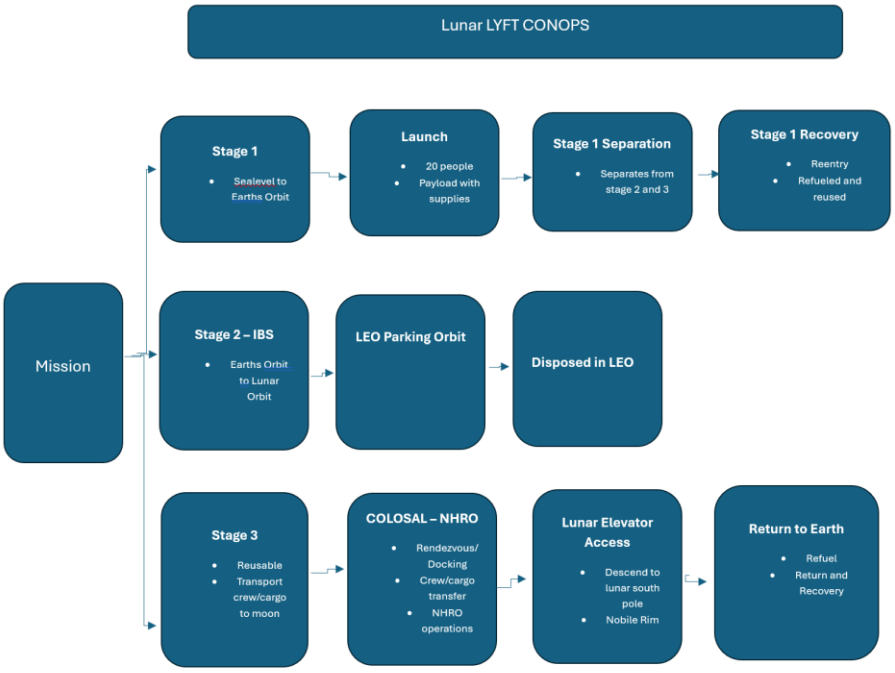



*Figure 1 above is the Lunar Lyft CONOPS.*

# Conceptual Design

## Stage 1 & 2 Vehicle

*Table 1: Trade study for stage 1 vehicle.*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* | *Model 4* |
|---|---|---|---|---|---|
| **Safety, failure** | 10 | 9 | 9 | 6 | 1 |
| **Budget Limit** | 1 | 7 | 7 | 7 | 2 |
| **Vehicle Reusability/Reliability** | 9 | 9 | 7 | 3 | 1 |
| **Complexity** | 5 | 4 | 3 | 7 | 2 |
| **Payload capacity (mass)** | 8 | 5 | 5 | 7 | 10 |
| **System Efficiency (power input vs output)** | 9 | 7 | 8 | 6 | 10 |
| **Total** | | ***301*** | ***287*** | ***239*** | ***201*** |

**Concept Explanation**

- **Model 1 – Hydrocarbon Liquid Propellant Engine:** Liquid engines offer a throttleable propulsion method utilizing relatively fewer toxic materials than the other engine types considered while providing a high specific impulse based on.
- **Model 2 - Hybrid Solid and Liquid Engine:** The hybrid engine is a simpler design than full liquid engines: one liquid turbopump with a combustion chamber containing the solid fuel. This model is throttleable but offers a lower thrust and efficiency over time, based on the solid component's regression rate, than classic liquid bipropellant engines.
- **Model 3 - Solid Engine:** Solid engines are a single-use, relatively simplistic design that contains no moving parts. Solid propellants are easily stored for long periods of time. Once it ignites, it burns non-stop at a high temperature and pressure, offering no throttle ability.
- **Model 4 - Nuclear Thermal Engine:** Nuclear thermal propulsion heats a liquid propellant, such as hydrogen, to a high-pressurized gas that produces almost twice the specific impulse as the average standard liquid engine. However, this model is primarily used in core staging for long missions, rather than initial staging.

**Selection Process and Results**

The commercialization of this mission concept dictates that safety and reliability, was weighed the most heavily to determine stage one rocket engine. Vehicle reusability/reliability includes the ability to recreate the engine configuration while also accounting for how much research and testing has been publicized on each model. Payload capacity and system efficiency encapsulate the specific requirement that 20 people, along with sufficient supplies, can be carried by this booster.

Overall, a liquid engine will perform best for the scope of this mission. Safety was the most important factor considered since this mission, will civilians aboard, must be performed many times to keep the cislunar habitations functioning. Liquid propellant rocket engines are the most used for initial staging. This builds credibility that is cost-effective and reliable, and therefore safe. The Falcon 9 booster, the world's first reusable orbital-class rocket, has successfully landed 643 times out of the 690 missions it has completed. The manufacturing of this booster over ten years ago sparked a new era of spaceflight, in which more reusable boosters with reentry and landing capabilities have been researched and refined.

## Stage 1 Trajectory

The stage 1 flight path begins at the ground; the first stage, Stage 1A and Stage 1B run parallel. Stage 1A is jettisoned at around 127 seconds while Stage 1B continues carrying the spacecraft out of the mesosphere.

## Stage 1 Fuel

*Table 2: Trade study for the stage 1 fuel ideas.*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* | *Model 4* |
|---|---|---|---|---|---|
| **Specific Impulse/Performance** | 10 | 8 | 9 | 8 | 6 |
| **Fuel Density** | 9 | 10 | 6 | 2 | 8 |
| **Storage Requirement** | 7 | 9 | 4 | 3 | 8 |
| **Operation Simplicity** | 6 | 8 | 5 | 3 | 8 |
| **Technology maturity** | 9 | 10 | 8 | 10 | 6 |
| **Refueling Complexity** | 5 | 8 | 5 | 3 | 8 |

| | | | | | |
|---|---|---|---|---|---|
| **Combustion Efficiency** | 7 | 8 | 9 | 10 | 7 |
| **Total** | | ***539*** | ***434*** | ***376*** | ***435*** |

**Concept Explanation**[1]

- **Model 1 - RP1:** It is a refined kerosene that stays liquid at room temperature, therefore does not need cryogenic handling Highest overall score. It leads to fuel density, it has
- s for mining. It also has a mixture of illuminated and non-illuminated areas, producing solar energy potential. However, terrain can potentially make it difficult for installation. Resources in the area are believed to be mostly water. In comparison to other models, terrain may be difficult for safety when landing.
- **Model 4 - Mons Mouton:** In comparison to other models, this concept has the best area for lunar base installations. Its terrain allows for water extraction, mining, safe landing of space vehicles, and potential long term storage areas.

**Selection Process and Results**

All models were selected based on criteria needed for crewed lunar base activity. Criteria such as mining possibilities, access to water resources, and easier access from NRHO were weighed higher. Terrain for vehicle landing and access was also considered communication and solar power availability. Overall, Noble Rim is the best model due to it having the best balance of all necessary criteria. Four different areas in the south pole were chosen, the Noble Rim Region, Malapert Massif region, de Gerlache Rim Region, and Mons Mouton region. Other Criteria for selecting these areas also include communication availability and access from the NHRO. Out of all the models presented, the trade study makes Noble Rim the best balance for all.

## Stage 2 Trajectory

*Table 3: Trade study for the trajectory of stage 2.*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* | *Model 4* |
|---|---|---|---|---|---|
| **Budget Limit** | 1 | 1 | 4 | 5 | 5 |
| **Vehicle Reusability** | 6 | 1 | 7 | 3 | 5 |
| **Propellant Mass Allocation** | 10 | 8 | 3 | 7 | 7 |
| **Allowable Payload Mass** | 10 | 7 | 4 | 5 | 6 |
| **Complexity** | 5 | 5 | 4 | 4 | 3 |
| **Stage Recovery Time** | 6 | 1 | 8 | 5 | 5 |
| **Delta-V Efficiency** | 8 | 9 | 3 | 5 | 6 |
| **Failure likeliness** | 9 | 6 | 4 | 3 | 4 |
| **Total** | | ***314*** | ***244*** | ***260*** | ***294*** |

**Concept Explanation**

- **Model 1**: Stage 2 jettisons after it performs TLI and does not get recovered. Considering the assumption of a very high budget, this is the most efficient configuration as no mass would have to be allocated to structure or propellant for recovery. However, operation costs would compound over the mission timeline as new stage 2 boosters would have to be manufactured for every launch.

[1]Refer to this concept explanation for all fuel trade studies

- **Model 2**: Stage 2 jettisons pre-TLI and performs a retro burn (large amount of fuel necessary) to quickly fall back to Earth's surface for recovery. This concept decreases the allowable payload mass along with extra mass being necessary for a recovery system, which is unfeasible for the mission requirements.
- **Model 3**: Stage 2 jettisons during TLI and extra propellant are used to enter a free return trajectory back to Earth using the moon's gravity. Extra propellant decreases the allowable payload mass contradicting mission requirements. The free return trajectory adds more complexity and has a high chance of failing unless extra propellant is onboard stage 2 during this return trajectory. The recovery period would take a couple of weeks to a month. If stage 2 in this configuration were to be reusable, extra mass would be allocated to a recovery system, further decreasing efficiency.
- **Model 4**: Stage 2 jettisons pre-TLI and enters a highly elliptical orbit around Earth with the apogee at ~10,000 km. Stage 2 performs retro burn after jettison to decrease apogee to ~5,000 km. This decreases the orbital period of the second stage allowing atmospheric drag and orbital decay to allow recovery of stage 2 in about 1 or 2 weeks. Since stage 2 is jettisoned, the required propellant mass to perform the retro-burn is decreased: slightly decreasing the allowable payload mass. This option does have some increase in complexity but is the most feasible real-world option for having a recoverable stage 2 while maintaining an efficient mass distribution.

**Selection Process and Results**

The models ranked above are consistent with their concept explanations. Most noticeably, Models 1 and 4 have the highest scores, with its propellant mass allocation and allowable payload mass being the difference in the total. Model 1 is the most mass efficient option and is used for this mission. Stage 2 will be in a LEO of ~500 km before jettisoning and remaining in orbit.

## Stage 2 Fuel

*Table 4: Trade study for Stage 2 fuel options.*

| Evaluation Criterion | Weight | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|---|
| **Specific Impulse/Performance** | 10 | 7 | 8 | 10 | 6 |
| **Fuel Density** | 8 | 10 | 6 | 2 | 8 |
| **Storage Requirement** | 7 | 9 | 6 | 3 | 8 |
| **Operation Simplicity** | 6 | 8 | 5 | 3 | 8 |
| **Technology maturity** | 10 | 10 | 8 | 10 | 6 |
| **Refueling Complexity** | 9 | 8 | 5 | 3 | 8 |
| **Combustion Efficiency** | 8 | 7 | 8 | 10 | 7 |
| **Total** | | 489 | 389 | 362 | 416 |

**Concept Explanation**[2]

**Selection Process and Results**

- **Model 1 - RP1:** It wins Stage 2 because it is the most practical fuel for a Moon-bound stage. Density is 10 (about 0.8 g/cc, so tanks stay small), storage is 9 (an ambient liquid, so only the oxidizer is cryogenic), simplicity and refueling are 8 (no cryogenic fuel system, easy transfer), and technology maturity is 10 from decades of Atlas, Soyuz and Falcon 9 flights. Its Isp is 7

[2] Refer to Stage 1 Fuel

because kerolox gives only about 330-350 s in vacuum; combustion efficiency is 7 because fuel-rich kerosene leaves some soot and deposits. Maturity (10), refueling (9) and density (8) carry heavy weights, so these strengths outweigh the weaknesses.

- **Model 2 – Methane:** Methane has the strongest performance among non–selected fuels. Isp is 8 (about 360-380 s), combustion efficiency is 8 (it burns cleanly), and maturity is 8 (Raptor and BE-4 fly, but with little in-space history). It is held back by being cryogenic at about -162°C. Storage is 6. Simplicity and refueling are 5 (cryogenic handling and transfer), and density is 6 (about 0.42 g/cc, so tanks are larger than RP1's). With refueling weighted at 9, those penalties keep it well behind RP1.
- **Model 3 – Hydrogen:** Hydrogen is the best performer and the worst to manage combustion efficiency is 10, maturity is 10 (RL10, RS-25 and Centaur), and Isp is 9 (about 450 s, the highest of the four). Density is 2 (about 0.07 g/cc, so large tanks), storage is 3 (about 20 K, with boil-off and heavy insulation), and simplicity and refueling are 3 (venting, leak management, and the hardest propellant to transfer). Stage 2 weights refueling at 9, density at 8 and storage at 7, so those penalties outweigh its efficiency and leave it last.
- **Model 4 – Ethanol:** Ethanol is easy to live with and low in performance. Density, storage, simplicity and refueling are all 8 (about 0.79 g/cc, an ambient liquid with benign handling), and combustion efficiency is 7 (it burns cleanly with no soot, but at a lower flame temperature than RP1). Its Isp is 6 (about 290-310 s, the lowest of the four), its maturity is 6 (V-2 and Redstone, with little modern use). Isp and maturity both carry a weight of 10, which keeps it behind RP1. It ends one point behind methane, so treat second and third as a tie.

## Stage 3

It is a transportation vehicle designed to support human and resource delivery to cislunar space. It will transport 20 passengers along with essential resources to a Near-Rectilinear Halo Orbit (NRHO), where the resources can support in-space operations. After completing its mission in NRHO, Stage 3 will return to Earth using a Trans-Lunar Injection (TLI) trajectory.

## Stage 3 Fuel

*Table 5: Trade study for Stage 3 fuel options.*

| Evaluation Criterion | Weight | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|---|
| **Specific Impulse/Performance** | 10 | 7 | 8 | 10 | 6 |
| **Fuel Density** | 4 | 10 | 6 | 2 | 8 |
| **Storage Requirement** | 6 | 8 | 6 | 3 | 9 |
| **Lunar TLI** | 10 | 2 | 6 | 10 | 2 |
| **Operation Simplicity** | 5 | 8 | 6 | 4 | 9 |
| **Technology maturity** | 8 | 10 | 8 | 9 | 7 |
| **Refueling Complexity** | 6 | 8 | 6 | 3 | 9 |
| **Combustion Efficiency** | 8 | 7 | 9 | 10 | 6 |
| **Propellant Stability** | 8 | 9 | 6 | 5 | 9 |
| **Lunar NRHO** | 10 | 7 | 8 | 10 | 6 |
| **Total** | | **544** | **530** | **556** | **501** |

**Concept Explanation**[3]

**Selection Process and Results**

- **Model 1 - RP1:** It is the closest challenger, 12 points behind hydrogen. It scores 10 on both density and technology maturity, 9 on propellant stability, and 8 on storage, simplicity, and refueling. As a dense, room-temperature liquid with decades of flight heritage, it needs no cryogenic systems and has no boil-off during coast or loiter. It loses because kerosene cannot be produced from lunar water. Its Isp, combustion efficiency, and NRHO scores (7 each) also count for more in a delta-v-limited round trip, and the reduced weights on density and storage shrink its biggest advantages.
- **Model 2 - Methane:** It has the best balance of performance and handling among the non-selected fuels. It scores 9 on combustion efficiency, 8 on Isp, NRHO, and maturity, and burns cleanly for reuse. It is also cryogenic at about -162°C, so storage, stability, simplicity, and refueling all score 6, and its density (6) requires larger tanks than RP1. That puts 26 points behind hydrogen.
- **Model 3 - Hydrogen:** It leads to pure performance, scoring 10 on Isp, combustion efficiency, Lunar TLI, and NRHO, with strong maturity (9) from upper stages like Centaur. It is the only fuel that mined lunar water supplies directly since electrolysis yields hydrogen and oxygen. It is extremely low density (2) demands large, insulated tanks, and its storage (3), refueling (3), simplicity (4), and stability (5) reflect boil-off and difficult handling at about -253°C. It wins here because a reusable stage refueled at a lunar depot does not launch its propellant from Earth, so those penalties count for less, and delta-v to NRHO and back counts for more.
- **Model 4 - Ethanol:** It is the easiest fuel to live with, scoring 9 on storage, simplicity, refueling, and stability, with good density (8). It is a non-toxic, room-temperature liquid with no boil-off. However, it has the lowest Isp and combustion efficiency (6 each) because of its oxygen and water content, and it scores 6 on NRHO, and it finishes last at 501.

## Stage 3 Trajectory

*Table 6:Trade study for the potential lunar orbits for COLOSAL: Near Rectilinear Halo (NRHO), Distant Retrograde (DRO), Low Lunar (LLO), and Projected Circular (PCO).*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* | *Model 4* |
|---|---|---|---|---|---|
| **Communication** | 9 | 10 | 9 | 5 | 8 |
| **Fuel usage** | 3 | 10 | 9 | 5 | 7 |
| **Stability** | 5 | 10 | 9 | 4 | 7 |
| **Accessibility to Earth** | 6 | 8 | 7 | 3 | 5 |
| **Accessibility to moon surface** | 7 | 7 | 6 | 10 | 9 |
| **Complexity** | 4 | 7 | 4 | 9 | 8 |
| **Total** | | ***295*** | ***253*** | ***204*** | ***286*** |

**Concept Explanation [12]**

[3] Refer to Stage 1 Fuel

**Model 1 – NRHO (Near Rectilinear Halo Orbit):**

- Orbital period: 6-8 days
- Lunar Surface Access
  - Time to lunar surface: 12 hours
  - Delta-u: 730 m/s

**Model 2 - DRO (Distant Retrograde Orbit):**

- Orbital period: roughly 14 days
- Lunar Surface Access
  - Time to lunar surface: 4 days
  - Delta-u: 830 m/s

**Model 3 - LLO (Low Lunar Orbit):**

- Orbital period: roughly 2 hours
- Lunar Surface Access
  - Time: less than an hour
  - Delta-u: 0-800 m/s depending on inclination

**Model 4 - PCO (Projected Circular Orbit):**

- Orbital period: roughly 11 hours
- Lunar surface access
  - Time: less than 5 hours
  - Delta-u: 700 m/s

**Selection Process and Results**

The final orbit for the cislunar trip, and the location of the space station COLOSAL, was analyzed by comparing different lunar orbits. Communication, including both COLOSAL to earth and COLOSAL to Tranquility, was weighed the most heavily since it influences the safety of the crew and mission. Fuel usage and stability were weighed similarly since they both depend on how much power the telemetry systems use once the station and 3rd stage are in orbit. Stability, however, was deemed more important since the comfortability of people staying in the station depends on how much wobble COLOSAL has in orbit. Fuel usage was the least important criteria since the mining and research facility would process and manufacture fuels used in the mission. Accessibility to the moon was measured by estimating the time and delta-u for a Hohmann transfer from each orbit to LLO which would then decay into the lunar surface. This factor was significant since the functionality of the cislunar habitation systems relies on the ability to make frequent trips back and forth from the bases to the station to carry people and supplies and refuel vessels.

## Lunar Bases

*Table 7: Trade study for lunar base concepts.*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* | *Model 4* |
|---|---|---|---|---|---|
| **Water availability** | 10 | 9 | 7 | 9 | 9 |
| **Solar Access for non-shadowed & shadowed areas** | 9 | 9 | 8 | 9 | 7 |
| **Communication** | 7 | 7 | 10 | 8 | 7 |
| **Landing safety for vehicles** | 9 | 7 | 7 | 7 | 9 |
| **Vehicles Access from NRHO** | 10 | 9 | 8 | 9 | 9 |
| **Mining potential** | 7 | 9 | 9 | 9 | 9 |
| **Total** | **52** | ***50*** | ***49*** | ***51*** | ***50*** |

**Concept Explanation**

- **Model 1 - Noble Rim Region:** This region is located near permanently shadowed regions, has water/ice access, and has the best balance of all criteria. Water/ice access may be used for fuel generation and systems such as life support. The Noble Rim region is also safer for vehicle access and general crewed missions due to its sufficient landing zones and terrain access that could be permanently lit. Shadowed/non-shadowed areas can also be used for mining, as the area is believed

to have resources such as metals and oxygen oxides. Permanently lit areas provide access to constant solar power.

- **Model 2 - Malapert Massif:** Due to high illumination, this region has high solar capabilities, good communication potential, and gives a great view of the earth. Terrain is not favorable for landing or mining due to variations in elevation.
- **Model 3 - deGerLache Rim:** This region also contains a mixture of shadowed and non-shadowed regions that are advantageous for mining. It also has a mixture of illuminated and non-illuminated areas, producing solar energy potential. However, terrain can potentially make it difficult for installation. Resources in the area are believed to be mostly water/ice. In comparison to other models, terrain may be difficult for safety when landing.
- **Model 4 - Mons Mouton:** In comparison to other models, this concept has the best area for lunar base installations. Its terrain allows for water extraction, mining, safe landing of space vehicles, and potential long term storage areas.

**Selection Process and Results**

All models were selected based on criteria needed for crewed lunar base activity. Criteria such as mining possibilities, access to water resources, and easier access from NRHO were weighed higher. Terrain for vehicle landing and access was also considered communication and solar power availability. Overall, Noble Rim is the best model due to it having the best balance of all necessary criteria. Four different areas in the south pole were chosen, the Noble Rim Region, Malapert Massif region, de Gerlache Rim Region, and Mons Mouton region. Other Criteria for selecting these areas also include communication availability and access from the NHRO. Out of all the models presented, the trade study makes Noble Rim the best balance for all.

## Vehicle for Lunar Surface to Tranquility

*Table 8: Trade study for picking a vehicle that transports from the lunar surface to Tranquility.*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* | *Model 4* | *Model 5* |
|---|---|---|---|---|---|---|
| **Safety** | 7 | 5 | 5 | 4 | 6 | 9 |
| **Sustainability: dependency on Earth for resupply** | 6 | 5 | 1 | 6 | 2 | 8 |
| **Reusability** | 10 | 8 | 1 | 8 | 10 | 10 |
| **South-pole-to-NRHO compatibility** | 10 | 8 | 6 | 8 | 10 | 7 |
| **TRL** | 6 | 8 | 3 | 4 | 5 | 6 |
| **Human Readiness** | 10 | 7 | 6 | 6 | 3 | 5 |
| **Travel Time** | 4 | 8 | 8 | 9 | 5 | 7 |
| **Departure Availability and Turn Around Time** | 2 | 7 | 3 | 7 | 5 | 6 |
| **Propellant/Energy Demand** | 5 | 5 | 5 | 2 | 10 | 8 |
| **Maintenance Requirements** | 6 | 5 | 1 | 5 | 9 | 7 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Total** | | ***364*** | ***258*** | ***398*** | ***446*** | ***489*** |

**Concept Explanation**

- **Model 1 - Reusable Chemical Single-Stage Shuttle**: A single stage vehicle that is refueled and serviced between missions uses chemical rocket engines to transport crew between the lunar surface and orbit. Stecklein's Lander Conceptual Design [1] and Komar's Lunar and Mars Ascent and Descent/Entry Crew Abort Modes for the Hercules Single-Stage Reusable Vehicle [2] are the basis for Model 1.
- **Model 2 - Partially Reusable Two-Stage Lander**: A descent stage lands the spacecraft and remains on the moon with a separate ascent stage returns to passengers to the hotel. The ascent stage is refueled, services and attached to a new descent stage in orbit. Howard's JUMP concept explores this idea thoroughly [3].
- **Model 3 - Nuclear Thermal Single-Stage Shuttle**: A reusable vehicle uses a reactor to heat the mined hydrogen and use it for thrust for ascent and descent. Nikitaeva explores this idea thoroughly in In Situ Propellant Alternatives for a Lunar Ascent/Descent Vehicle [4].
- **Model 4 - Rotating Orbital Tether**: A rotating cable attached to a weight briefly matches its tip velocity to the lunar surface, picking up a passenger vehicle. The tether transfers its momentum to the vehicle, putting it on an orbital trajectory to be put in orbit and attach to Tranquility. This idea is investigated by Hoyt in a very intriguing explanation [5].
- **Model 5 - Surface-Anchored Tether**: A cable connects the lunar surface to a counterweight beyond the Earth to Moon L1 or L2 balance points, allowing a solar powered climber to carry passengers between the surface and space. Pearson et al. describes this architecture quite thoroughly and is the supporting documentation for this model [6]. Other supporting documentation includes Eubanks and Radley's Scientific Return of a Lunar Elevator [7], Shi et al. Cargo-Transfer Libration Control for a Partial Elevator [8], and Liu et al. analyzing the statics and dynamics of an L1-side anchored tether [9].

**Selection Process and Results**

The trade study method was used to evaluate five vehicle models to find the ideal method for transporting passengers from the Lunar Base to Tranquility, the hotel in NRHO, and vis versa. The highest weighted criteria: reusability, south-pole-to-NRHO compatibility, TRL (technology readiness level/current amount of supporting research) was prioritized because of the main goals this vehicle needed to solve. Each value given was thought out utilizing respective information from references.

The clear winner for this trade study is Model 5: Surface-Anchored Tether (Figure2), exceeding the second highest concept by more than forty points. Both tether concepts scored favorably, but Model 5's better safety and sustainability set it further apart. The tether models offered an unprecedented concept that limits propellant need but scored less favorably in human readiness and departure availability/turnaround time.

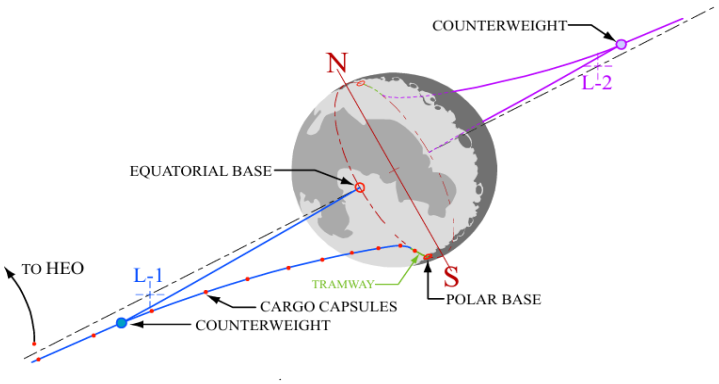



*Figure 2 - STAR inc. space elevator concept about L1 and L2.*

## Re-Entry and Return Trajectory

*Table 9: Trade study for re-entry and return trajectory method for Stage 3.*

| Evaluation Criterion | *Weight* | *Model 1* | *Model 2* | *Model 3* |
|---|---|---|---|---|
| **Thermal Protection** | 10 | 5 | 7 | 7 |
| **Crew Acceleration (Safety)** | 10 | 3 | 8 | 9 |
| **Landing Accuracy and Accessibility** | 9 | 3 | 7 | 9 |
| **Guidance and Control Simplicity** | 6 | 10 | 7 | 4 |
| **Technical Maturity** | 8 | 8 | 10 | 8 |
| **System Mass** | 7 | 9 | 7 | 7 |
| **Total** | | ***294*** | ***384*** | ***378*** |

**Concept Explanation**

- **Model 1 - Ballistic Entry:** Uses atmospheric drag with little aerodynamic lift to slow the spacecraft. It offers simple guidance and relatively low system mass, but limited landing control and higher crew acceleration resulted in the lowest score of **294**.
- **Model 2 - Lifting Direct Entry:** Uses aerodynamic lift and bank-angle control during a single atmospheric entry to manage deceleration and guide the spacecraft toward its landing site. It balances crew safety, landing accuracy, thermal protection, and control complexity, earning the highest score of **384**.
- **Model 3 - Skip Entry:** Uses lift to rise toward a thinner atmosphere before descending again. It offers lower crew acceleration and greater landing-site accessibility, but more complex guidance reduced its overall score to **378**.

**Selection Process and Results**

Although lifting direct entry achieved the highest weighted score of **384**, skip entry scored closely at **378**, a difference of approximately **1.6%**. The team selected skip entry because its advantages in reducing crew acceleration and increasing landing site accessibility were considered more valuable for the mission than its added guidance complexity.

# Performance Analysis

The performance analysis below was conducted considering these requirements in Table 11.

*Table 11: WBS and mission requirements.*

| WBS# | Element | L1 | L2 | Requirements |
|---|---|---|---|---|
| 1 | Mission 1 | X | | • Mission is completed safely and in a timely manner.<br>• Minimal cost for resupplying stages<br>• Must turn a significant profit |
| 1.1 | Stage 1 | | X | • Must have landing system<br>• Must be reusable and refuellable<br>• Must produce enough thrust to carry payload to mesosphere<br>• Must have jettison and recovery system |
| 1.2 | Stage 2 | | X | • Enough thrust and fuel to get to LEO parking orbit<br>• Must have jettison system<br>• Must have high structural efficiency |
| 1.3 | Stage 3 | | X | • Must reengage Earth's atmosphere and return with minimal damage<br>• Life support systems in payload area capable of safely transporting humans<br>• Storage space for tools, food, and amenities<br>• Must have docking system to rendezvous with COLLOSAL station |

*Table 12: Performance Analysis for stages 1 through 3.*

| | Falcon Heavy* S1-A | Falcon Heavy* S1-B | Stage 2 Falcon 9* S2 | Stage 3 (LOX/LH2) S3 | Total |
|---|---|---|---|---|---|
| | *3 cores: liftoff to side sep* | *Center core to MECO* | | 3 × RL10C-X | *Total Δu, all phases* |
| M0 (kg) | 1,403,919.2 | 426,832.8 | 147,219.2 | 35,719.2 | |
| Mp (kg) | 835,718.4 | 256,413.6 | 107,500.0 | 20,472.0 | |
| Ms (kg) | 141,368.0 | 23,200.0 | 4,000.0 | 12,000.0 | |
| ML (kg) | 426,832.8 | 147,219.2 | 35,719.2 | 3,247.2 | |
| Mb (kg) | 568,200.8 | 170,419.2 | 39,719.2 | 15,247.2 | |
| R | 2.47081525 | 2.50460511 | 3.70649963 | 2.34267275 | |
| epsilon | 0.14468321 | 0.08297164 | 0.03587444 | 0.36954915 | |
| lambda | 0.43684243 | 0.52650944 | 0.32035157 | 0.10000000 | |
| Isp (s) | 290 | 305 | 348 | 461 | |
| ueq (m/s) | 2,844.90 | 2,992.05 | 3,413.88 | 4,522.41 | |
| Du (m/s) | 2,573.349 | 2,747.094 | 4,472.483 | 3,849.894 | 13,642.820 |
| mdot (kg/s) | 6,597.70 | 2,749.04 | 270.78 | 72.97 | 9,792.926 |
| tb (s) | 126.7 | 93.3 | 397 | 280.6 | |
| T (N) | 18,769,787.2 | 8,225,266.0 | 924,413.4 | 330,000.0 | |
| (kN) | 18,769.787 | 8,225.266 | 924.413 | 330.000 | |
| g0 | 1.36285053 | 1.96436938 | 0.64007778 | 0.94176644 | |
| gbo | 3.36735188 | 4.91996961 | 2.37244804 | 2.20625057 | |

From table 12, stage 1 and 2 carry stage 3 up to LEO, where stage 3 performs a TLI to a lunar NRHO. Propellant and structural masses are loosely based off the falcon heavy data found from sources [17] and [18]. Assuming the delta V required for LEO is 9400 m/s, the summed stage 1 and 2 delta V's produce 9792.926 m/s, with 292.926 m/s to spare, which closes. Next a TLI is performed with stage 3, using 3 RL10C-X engines, stage 3 can produce 3849.894 m/s delta V. Being conservative, the delta V required for a TLI is about 3250 m/s, and NRHO is about 830 m/s. Using the left-over delta V from LEO orbit, the total delta V produced by stage 3 is 4124.82 m/s which closes the conservative 4080 m/s required for this

mission. If circumstances arise, stage 1 can use all its fuel to produce a total of 14143.357 m/s delta V which is more than enough to complete the mission; however, stage 1 would no longer be recoverable.

Stage 1 is broken down into S1-A and S1-B. This is because the optimal way of utilizing the 2 side boosters and center core is to manipulate the throttling of the center core. Table 12 shows during S1-A, the center core is throttled 40% while the two side stages are fully throttled. S1-B is after the side stages that uses all but 12% of its fuel; this is because the rest of the fuel will be used to recover the side boosters back to Earth to be reused and refueled. During this time, the center core is fully throttled 100% and continues until the main engine is cut off, where it separates with stages 2 and 3. Representing and calculating the data this way increases the delta v by 2000 m/s compared to a lumped stage 1 design. When calculating, the throttled center core must be accounted for. Breaking down each booster, we assume from space X data that each core holds about 395700 kg of propellant. Considering 12% leftover fuel for landing, we can use the equation:

$$B = P(1 - f) \quad [1]$$

Where B is the propellant each booster burns during takeoff, P is the total propellant per core, and f is the percentage of leftover fuel needed for recovery. Using 12%, the propellant burn of the side boosters is 348216 kg. Using published Isp and thrust values for a Merlin 1D engine, we can calculate the mass flow rate for the engine. According to SpaceX data, the center core is throttled between 40 to 50% during the side booster burn. Multiplying the mass flow rate by 9 for the 9 engines in the center core and by 40% gives us a center core propellant mass of 139286.4 kg. Adding the values calculated gives us the propellant mass shown in table 11. Structural mass accounts for left over propellant landing of the side boosters and the structural mass of each core. Other masses and ratios are calculated the same for the two stages. Stage 1-B is calculated with the remaining propellant left over from stage 1-A burn without the mass of the two booster stages.

The structural and propellant masses for stage 2 were taken directly from the mass values of the Space X Falcon 9 second stage. The required structural and propellant masses for stage 3 were based on the SpaceX Dragon and scaled to meet the 20-person passenger count. The payload mass for stage 3 is the estimated life support required to sustain 20 passengers for the 6-day trip to the Lunar Space Station. This includes food, water, luggage, oxygen, and medical supplies for emergencies.

The total masses for each stage were calculated by:

$$M_0 = M_L + M_S + M_P \quad [2]$$

Where $M_L$ is the payload mass, $M_S$ is the structural mass, and $M_P$ is the propellant mass. The burnout masses for each stage were calculated by:

$$M_b = M_0 - M_P \quad [3]$$

The structural coefficients for each stage were calculated by:

$$\epsilon = \frac{M_S}{M_P + M_S} \quad [4]$$

And the payload ratios for each stage were calculated by:

$$\lambda = \frac{M_L}{M_0 - M_L} \quad [5]$$

For the mass fractions, the following equation was used:

$$R = \frac{M_0}{M_b} \quad [6]$$

The equivalent exhaust velocities for the engines in all stages were calculated using the following equation:

$$u_{eq} = Isp * 9.81 \frac{m}{s^2} \quad [7]$$

The Isp values were found from data on the rocket engines used in each stage. For S1-A, the Isp value was for the Merlin 1D engine at sea level. For S1-B, the Isp value was for the Merlin 1D engine in the upper atmosphere. For stage 2, the Isp was for the Merlin 1D engine in a vacuum. And for stage 3, the Isp value was for the RL10C-X engine operating in a vacuum.

The delta-u values for all stages were then computed by:

$$\Delta u = u_{eq} \ln(R) \quad [8]$$

The burn out time for stage 2, 397s, is based on SpaceX data on the vacuum optimized Merlin 1D engine. From this burnout time, the mass flow rate of the second stage is calculated by dividing the stage 2 propellant mass by the burnout time as follows:

$$\dot{m}_2 = \frac{M_{P2}}{t_{b2}} \quad [9]$$

The stage 2 thrust is then calculated by multiplying this mass flow rate by the equivalent exhaust velocity of the stage 2 engine as follows:

$$T_2 = \dot{m}_2 * u_{eq2} \quad [10]$$

The thrust value for stage 3 is based on an engine configuration consisting of 3 RL10C-X rocket engines. Each individual engine produces a thrust of 110,000 N, bringing the total thrust to 330,00 N. The required mass flow rate for this configuration is computed by:

$$\dot{m}_3 = \frac{T_3}{u_{eq3}} \quad [11]$$

The burnout time for this st age can then be calculated as follows:

$$t_{b3} = \frac{M_{P3}}{\dot{m}_3} \quad [12]$$

The G-force experienced at the initial burn of each stage is then calculated by:

$$g0 = \frac{T}{M_0 \left(9.81 \frac{m}{s^2}\right)} \quad [13]$$

And similarly for the G-force experienced at the burnout of each stage:

$$gb0 = \frac{T}{M_b \left(9.81 \frac{m}{s^2}\right)} \quad [14]$$

Stage 3 payload mass is broken down using NASA data [12]. A broken-down form of each component weight times the amount of people carried on board is shown below:

*Table 12: Payload Mass of Stage 3 using NASA data*

| | Rate | x 20 people and 6 days | |
|---|---|---|---|
| Food | 2.39 | 286.8 | kg |
| Wipes and Gloves | 0.2 | 24 | kg |
| hygine kit | 2.4 | 48 | kg |
| Personal Storage | 5 | 100 | kg |
| Operational Supplies | 2.5 | 50 | kg |
| Health Care Cons. | 0.09 | 10.8 | kg |
| Waste and Hygine | 0.3 | 36 | kg |
| LiOH Canister | 1.75 | 210 | kg |
| Oxygen Metabolic | 0.89 | 106.8 | kg |
| Water (82kg crew) | 2.79 | 334.8 | kg |
| 82kg Crew | 82 | 1640 | kg |
| Luggage (20 kg limit) | 20 | 400 | kg |
| | | 3247.2 | kg |

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